

GPT-6 Luna across every effort level

VulcanBench Frontier v4 | Code quality protocol v3.16 | September 2026

Abstract

GPT-6 Luna ran the 23-task VulcanBench Frontier v4 suite through Codex CLI 0.155.0 on a ChatGPT Pro subscription on September 25 to 28, 2026, once per task at each of the five effort levels Codex offers for it, 115 runs in all. Code quality carries 33% of the combined score and is judged for a named human reader by Muse Spark 1.3 (Meta) and Grok 4.6 (xAI) under the same frozen protocol as the other Frontier v4 reports, with a ground-truth intent-recovery probe. Over judged runs the combined score rises at every step, from 40.83 at Low to 45.72 at Medium, 57.35 at High, 78.79 at Extra-high and 81.42 at Max. GPT-6 Luna passes 0 of 23 tasks at Low and Medium, 2 at High, 11 at Extra-high and 12 at Max. Code quality stays between 65.84 and 71.51. Cost per priced task runs from \$0.010 at Medium to \$0.160 at Extra-high; the whole sweep prices at \$7.26.

Six runs hit the flat 3-hour task bound while still working: 2 at Extra-high and 4 at Max. They have no finished code to judge, so the standard combined score and Code quality at those levels cover 21 and 19 of 23 runs. Every combined score in this report is therefore given two ways: over judged runs, as for every other board column, and with each timeout counted as 0 over all 23 runs, which gives 71.94 at Extra-high and 67.26 at Max. Tasks passed always counts timeouts as failures.

Table 1. Combined score two ways, Code quality, runtime and tasks passed by effort

Effort	Combined (33%)	SE	Timeouts as 0	Combined (20%)	Code quality	Min/task	Judged	Passed
Low	40.83	2.04	40.83	44.00	65.84	5.6	23	0/23
Medium	45.72	1.32	45.72	48.47	69.33	3.6	23	0/23
High	57.35	3.80	57.35	60.41	66.40	15.6	23	2/23
Extra-high	78.79	3.02	71.94	81.21	71.49	52.6	21	11/23
Max	81.42	2.88	67.26	83.83	71.51	60.5	19	12/23

Combined (33%) = 0.50 functional + 0.085 lint and complexity + 0.085 security + 0.33 Code quality, over judged runs. Timeouts as 0 counts each run stopped at the 3-hour bound as a combined score of 0 over all 23 runs. Combined (20%) applies the prior 50/15/15/20 profile to the same Code quality scores over judged runs. SE is one sample standard error across the cell's judged tasks, not judge uncertainty or a significance test. Passed counts all 23 runs. Runtime is solver wall-clock per task over every run in the cell, timeouts at their recorded 180 minutes; judging is excluded.

The Code quality protocol

Two failure modes motivated the reviewed score. The lint and complexity metric rewards compression: the maintainability index carries a lines-of-code term and per-function complexity stays low when each dense line does something different, so five statements on a line with names like *x* and *k* can outscore the same logic written for a person. And a large model reading compressed code pays almost nothing to parse it; when a frontier model was calibrated as a readability judge, it rated a squashed six-line function the same as its formatted copy. Since September 7, 2026 Code quality carries 33% of the combined score and lint and complexity and security carry 8.5% each. The prior profile is reported beside the new one on every table.

Protocol v3.16 is the same protocol applied to this population. Nothing in the rubric, controls, quirk keys, gates, repeats, seed, weights or judges changed from v3.7; both judges retook the calibration exam under v3.16 before any counted call. The population froze on September 28 with 109 rows, none missing, and six runs excluded as incomplete source runs: the six that hit the 3-hour bound.

The rubric

Every judge receives the same instructions, frozen by hash before any review. The prompt names the reader it scores for: an engineer who has never seen the code, reads it top to bottom without running it, and must make a correct change in one sitting. It tells the judge that its own ease at parsing dense code is not evidence of readability. Six dimensions are scored 0 to 4 in half steps against written anchors, each with an exact excerpt and a concrete consequence for that reader.

Sub-score	Dimension	What is assessed
Human readability	Naming	Identifiers say what things are in the task's domain
Human readability	Presentation	Statements per line, nesting, function length, how much the reader must hold at once
Human readability	Intent	Constants, thresholds and quirks explained by names or accurate comments that say why
Maintainability	Structure	Coherent responsibilities, no duplicated policy, no abstraction beyond the task's scale
Maintainability	Changeability	A named plausible change, every edit site traced, scored on locality and safety
Maintainability	Verifiability	Explicit state, failures that name what went wrong, seems to test one rule alone

The layers under the 33 points

Reviewed panel, 24 points. The six-dimension rubric above, averaged across both judges. **Intent recovery, 9 points.** Every task in this suite is a rewrite of a retired binary whose real behaviour departs from its written specification in documented ways. The judge sees only the specification and the code, never the issue text, and lists where the code departs from the specification; a separate call matches that list against the frozen answer key. The denominator is the quirks the submission actually passed tests for, so a functional failure is not punished twice; a submission that passed none has no intent-recovery score and its Code quality is the reviewed score alone. 14 GPT-6 Luna runs needed that rule (10 at Low, 3 at Medium, 1 at High). **Measured maintenance, designed for 12 points, not yet built.** Until it exists the pre-registered split above is in force.

Judges and calibration

The scored panel is Muse Spark 1.3 through the Muse CLI on its Standard tier, which does not train on prompts, and Grok 4.6 through the Cursor CLI at medium effort, the v3.7 pair under the same pinned binaries. Neither lab has a model on this board, so neither judge grades a relative. Every session is fresh, tools are disabled, the workspace is empty, model identity is checked per call from the CLI's own records, and solver labels are withheld.

The calibration exam

Before scoring a single submission each judge reviews ten held-out programs that implement the same ledger specification: clear, compressed, compressed then auto-formatted, verbose with duplicated policy, needlessly abstracted, misleadingly commented, narrated with a comment on every line, a documented legacy quirk, an embedded instruction to give full marks, and hidden module state. Each program is reviewed five times in a seeded order. Twenty gates fixed in advance check that the judge sees the construct; a judge may miss at most one gate by at most half a point.

Table 2. Calibration verdicts

Judge	Protocol	Calls	Result	Allowance	Failing gate
Muse Spark 1.3	code-quality-maintenance-v3.16	80	passed	used	g11_repeatability (short by 0.06)
Grok 4.6	code-quality-maintenance-v3.16	80	passed	used	g04_formatting_is_presentation (short by 0.10)

Both judges passed, each using the pre-registered one-gate allowance: Muse Spark 1.3 fell 0.06 short on the repeatability gate and Grok 4.6 fell 0.10 short on the gate that checks formatting is credited as presentation, both well inside the half-point bound. Every other gate passed for both.

Table 3. Control means on five of the ten calibration programs

Judge	Control	Naming	Present.	Intent	Struct.	Change.	Verif.
Muse	clear	4.00	4.00	3.40	4.00	4.00	3.60
Muse	compressed	1.10	1.90	1.10	3.90	2.50	3.00
Muse	formatted	1.10	2.80	1.00	3.90	2.50	3.00
Muse	misleading comments	3.80	4.00	0.20	4.00	2.50	3.60
Muse	hidden state	2.50	3.20	2.00	1.50	2.60	1.80
Grok	clear	3.70	3.50	3.10	4.00	3.90	3.50
Grok	compressed	1.40	1.60	1.10	3.00	2.50	2.90
Grok	formatted	1.60	2.40	1.70	3.20	2.60	3.00
Grok	misleading comments	3.50	3.30	1.10	3.50	2.40	3.00
Grok	hidden state	2.90	2.90	2.30	2.00	3.00	2.10

Full gate values and control means are in calibration.json.

Results in detail

Table 4. Code quality components at each effort level

Component	Low	Medium	High	Extra-high	Max
Code quality	65.84	69.33	66.40	71.49	71.51
Human readability	59.8	64.9	60.3	67.4	66.9
Maintainability	66.3	72.0	67.5	72.1	72.3
Intent recovery	80.2	72.8	73.8	76.1	76.7
Reviewed score	63.0	68.4	63.9	69.7	69.6
Rated by Muse Spark 1.3	62.8	67.4	63.0	68.0	68.5
Rated by Grok 4.6	63.3	69.5	64.8	71.5	70.6
Standard error of Code quality	2.24	1.78	1.50	1.02	1.52
Judged runs	23	23	23	21	19

Code quality sits between 65.84 and 71.51, with Grok 4.6 the higher rater at every level. Intent recovery at Low covers only the 13 Low runs that passed any documented departure, so it is not comparable with the other levels. Means cover judged runs.

Table 5. Four factors by effort, mean score out of 100 over judged runs

Effort	Functional	Lint, complexity	Security	Code quality
Low	7.54	80.41	100.00	65.84
Medium	14.92	80.98	100.00	69.33
High	40.30	79.91	100.00	66.40
Extra-high	79.73	80.34	100.00	71.49
Max	85.03	80.12	100.00	71.51

Table 6. Runs stopped at the 3-hour bound

Effort	Task	Minutes	Judged	Priced
Extra-high	depotcore	180.0	no	no
Extra-high	paddockcore	180.0	no	no
Max	cellarcore	180.0	no	no
Max	depotcore	180.0	no	no
Max	lodgecore	180.0	no	no
Max	paddockcore	180.0	no	no

Each trace shows continuous work to the bound, with no idle gap over four minutes. A timed-out run has no finished submission, so v3.16 excludes it from judging, as v3.15 excluded an incomplete Claude Opus 5.5 run. With four of 23 Max runs out of the judged set, all failures, the judged mean alone flatters the top levels; the owner's decision (harness decision log, September 28) is to publish both figures everywhere a score appears, with the judged figure as the headline cell value because it is computed the same way as every other board column.

Cost and tokens across effort levels

The 109 finished runs priced from their Codex receipts at list API rates, cache-aware, solver inference only, judging excluded. GPT-6 Luna ran on a ChatGPT Pro subscription, so these are API-equivalent estimates rather than bills. The six timeouts ended before Codex reported usage: they are unpriced, not \$0, so Extra-high and Max spend is understated. Medium is the cheapest and fastest level, and Extra-high the most expensive per priced task.

Table 7. API-equivalent cost, raw tokens and runtime by effort

Effort	Priced runs	\$/task	Level total	Tokens/task	Min/task
Low	23	\$0.028	\$0.64	1.65M	5.6
Medium	23	\$0.010	\$0.22	0.40M	3.6
High	23	\$0.052	\$1.19	2.56M	15.6
Extra-high	21	\$0.160	\$3.35	8.47M	52.6
Max	19	\$0.098	\$1.85	4.32M	60.5
Full sweep	109	\$0.067	\$7.26	366M	52.8 h

Rates checked 2026-09-25: GPT-6 Luna at \$0.10 input, \$0.01 cached input and \$0.50 output per million tokens. Codex receipts report input, cached input, output and reasoning tokens per run, so cached input is billed at the cache-read rate and the rest at the standard rate. The sweep stamped each run at these rates at run time; the export recomputed every priced run from its receipt and matched every stamp. Tokens are raw solver totals including cache reads, over priced runs. Runtime covers all 115 runs.

Limitations recorded with the estimates

- Codex receipts report input, cached input, output and reasoning tokens per run; cached input is billed at the cache-read rate and the rest of the input at the standard rate. Cache writes are free on this API and are not modelled.
- Standard tier, short-context rates. OpenAI bills prompts over 272K input tokens at a higher rate, but Codex keeps each request within its 272K context window, so no long-context premium applies and none is modelled.
- No Batch, Flex, Fast or priority pricing is applied.
- The six timed-out runs have no usage receipt. They are unpriced and excluded from cost and token means and totals; level means cover priced runs only, so extra-high and max spend is understated.
- The estimate covers the solver's exposed receipts, not an independently observed API invoice.
- The sweep stamped each run at run time at the published GPT-6 Luna list rates; the export recomputes every priced run from its receipt and matched every stamp.

How it was run, and the evidence

Codex CLI 0.155.0. Every other Codex column on the board ran on 0.153.4, which refuses GPT-6 Luna on a ChatGPT account before any work; 0.155.0 is the lowest release that serves it and was installed beside the global CLI for this sweep only. **Operator stop.** The first Extra-high pacecore run went silent after a Codex reconnect message; after about 88 idle minutes the operator stopped it, the harness re-queued it, and the retry is the counted run. **Judging alongside another sweep.** Judging ran on September 28, 00:32 to 14:32 PDT, alongside the GPT-6 Sol solver sweep at the owner's request; the GPT-6 Luna runs had finished. **One attempt per task and level;** effort labels are Codex's own.

Public files: `runs.json` and `runs.csv` with every run's factors, both judges' six-dimension sub-scores and intent recovery for each judged run, and the six timeouts; `groups.json` with both combined figures; `calibration.json` with every gate value and control mean; `judge-protocols.json` with the exact rubric, system text, probe and match instructions, schemas and weights; `economics.json` with cost and token aggregates, the rate table and pricing limitations; and `provenance.json` with source hashes and limits.

<https://github.com/morganlinton/VulcanBenchCOM/tree/main/assets/data/swe-v4-gpt6-luna-v316>

Withheld: raw prompts and responses; submitted patches and reconstructed sources; quirk answer keys (they describe hidden-test behaviour); reviewer session identifiers and usage receipts.

Recompute the published numbers

Each run's Code quality is $(0.24 \times \text{reviewed score} + 0.09 \times \text{intent recovery}) / 0.33$, where both terms are the equal mean of the two judges, or the reviewed score alone where the submission passed no quirk family. The combined score is $100 \times (0.50 F + 0.085 Q + 0.085 S + 0.33 C / 100)$ with F, Q, S on a 0 to 1 scale; the timeouts-as-0 figure uses 0 for each timed-out run. The export script recomputes every row from the frozen summary, re-prices every priced run from its receipt, checks both combined figures against the harness card tables, and refuses to write if any value differs.

What this report does not claim

No human rated anything; the scores are model judgment for a human reader, not human validation. Standard errors describe task sampling only. The measured-maintenance layer is unbuilt, so the 33 points are 24 reviewed plus 9 intent recovery. Runs were not repeated. The Extra-high and Max standard combined scores and Code quality are means over 21 and 19 of 23 runs. Hash checks bind the export to frozen files; they do not prove that judges were unbiased or that no training overlap exists.

Source hashes

Frozen artifact	SHA-256
v3.16/summary.json	a4fe3c64c43a61cc8327fce6612ff9f2...
v3.16/protocol.json	db301bf81db907d13e0ecff98699951...
v3.16/private-manifest.json	2a51cb811911b5d81b00f0f4a2340487...
v3.16/calibration-muse.json	e381ac093af8344f3ce359063ccb6797...
v3.16/calibration-grok.json	1681c1215e0c751aa46781aefed31600...
comparison.json	6cfc0dc8f703992a247d95d2132f1258...
gpt6-luna-v316-efforts.csv	8d26b78d6ffe8ffb3867e844bcc9d508...
gpt6-luna-v316-economics-efforts.csv	f77b93e5f604313bfafd76af3be49c89...

`provenance.json` also carries the hashes of the six timed-out runs' summaries.

Appendix. Per task at Low, Extra-high and Max

Task	Low CQ	Max CQ	Low combined	Extra-high combined	Max combined
blendcore	56.4	66.0	34.18	89.70	87.33
cellarcore	67.4	timeout	37.30	75.15	timeout
codeccore	57.3	73.1	34.45	54.74	69.49
depotcore	69.7	timeout	37.93	timeout	timeout
freightcore	76.0	81.8	40.68	79.79	81.28
granarycore	47.9	49.1	30.53	59.90	79.75
hedgcore	82.6	69.4	49.11	90.65	88.22
lodgecore	58.3	timeout	34.25	85.83	timeout
matchcore	65.2	75.8	36.74	77.78	90.00
metercore	80.2	65.9	42.16	89.16	75.99
pacecore	42.7	77.6	29.45	40.89	41.15
paddockcore	61.5	timeout	34.46	timeout	timeout
payrollcore	68.2	73.9	38.01	88.65	89.63
qlite	83.3	73.5	42.99	90.85	89.47
queuecore	65.9	75.4	37.21	70.45	77.74
quotacore	70.2	71.2	73.52	90.69	89.26
reflowcore	65.2	76.2	55.95	89.13	90.75
schedcore	49.2	70.8	31.20	88.35	76.03
settlecore	69.8	71.2	38.37	75.62	63.65
snapcore	66.7	71.5	47.60	64.54	89.26
stampcore	60.6	74.5	52.14	87.69	90.13
tallycore	77.1	72.5	41.11	87.24	89.41
vaultcore	72.9	69.4	39.78	77.71	88.49

Per-task values from runs.json. Task names are shortened for width. A timeout has no Code quality or standard combined score; in the timeouts-as-0 figure it counts as 0.

VulcanBench Frontier v4: GPT-6 Luna across effort levels

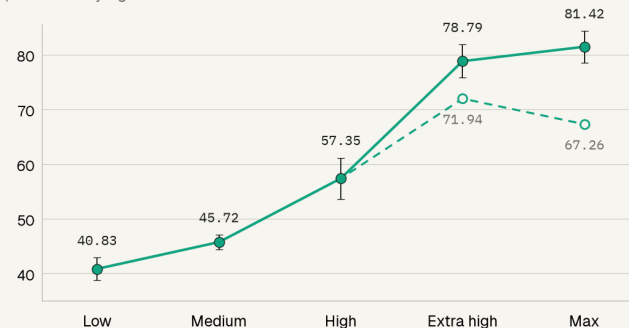
Combined score and runtime at every effort level, 23 tasks per effort. Code quality judged by Muse Spark 1.3 and Grok 4.6.

● GPT-6 Luna · Codex 0.155.0

n=23 runs per effort; judged n=21 at extra high, 19 at max

Combined score

/100 · solid: judged runs · dashed: timeouts counted as 0



Mean runtime

Minutes per task, all 23 runs · lower is better

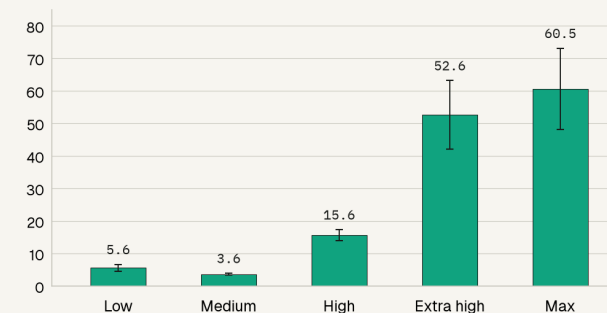


Table 1 | Scores at each effort level

Component	Low	Medium	High	Extra high	Max
/100; Code quality rows are the mean of both judges					
Combined score, judged runs	40.83	45.72	57.35	78.79	81.42
Combined score, timeouts counted as 0	40.83	45.72	57.35	71.94	67.26
Code quality	65.84	69.33	66.40	71.49	71.51
Human readability	59.8	64.9	60.3	67.4	66.9
Maintainability	66.3	72.0	67.5	72.1	72.3
Intent recovery	80.2	72.8	73.8	76.1	76.7
Rated by Muse Spark 1.3	62.8	67.4	63.0	68.0	68.5
Rated by Grok 4.6	63.3	69.5	64.8	71.5	70.6
Standard error of Code quality	2.24	1.78	1.50	1.02	1.52
Tasks passed, of 23	0	0	2	11	12
Runs stopped at the 3-hour bound	0	0	0	2	4

Six runs hit the flat 3-hour task bound while still working (extra high 2, max 4). They have no finished code, so they are not judged: the standard combined score averages the judged runs, as on every other board column, and the dashed line counts each timeout as 0 over all 23 runs. Tasks passed and runtime cover all 23 runs; a timeout counts as a failed task at 180 minutes. Extra high pacecore was restarted after an 88-minute Codex client stall; the retry is the counted run. Ran on Codex CLI 0.155.0, the first release that serves GPT-6 Luna on a ChatGPT plan.

VulcanBench Frontier v4: GPT-6 Luna across effort levels

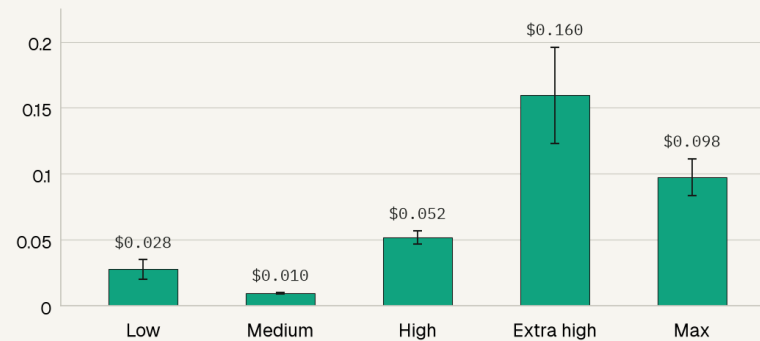
API-equivalent cost and token use at every effort level. 109 of 115 runs priced; solver inference only, judging excluded.

● **GPT-6 Luna · Codex 0.155.0**

priced n=23, 23, 23, 21, 19 from low to max

API-equivalent cost

USD per priced task at list prices · lower is better



Tokens per task

Millions of raw tokens, cache reads included · lower is better

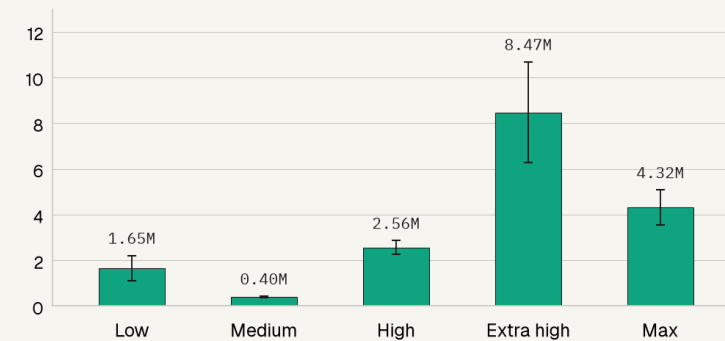


Table 1 | Cost, tokens, runtime and score at each effort level

Effort	Priced of 23 runs	\$/task	Level total API-equivalent	Tokens/task raw, millions	Minutes/task all 23 runs	Combined judged runs	Combined timeouts as 0
Low	23	\$0.028	\$0.64	1.65M	5.6	40.83	40.83
Medium	23	\$0.010	\$0.22	0.40M	3.6	45.72	45.72
High	23	\$0.052	\$1.19	2.56M	15.6	57.35	57.35
Extra high	21	\$0.160	\$3.35	8.47M	52.6	78.79	71.94
Max	19	\$0.098	\$1.85	4.32M	60.5	81.42	67.26
Full sweep	109	\$0.067	\$7.26	366M	52.8 h		

USD at list prices checked 2026-09-25, cache-aware (GPT-6 Luna \$0.10 input, \$0.01 cached input, \$0.50 output per million), solver inference only; subscription bills differ.

The six runs that hit the 3-hour bound (extra high 2, max 4) ended before Codex reported usage, so they are unpriced, not \$0: extra high and max spend is understated.

Runtime covers all 23 runs, timeouts at 180 minutes. Combined figures come from the score card: judged runs only, and with each timeout counted as 0 over 23 runs.

Whiskers are one task standard error. Tokens are raw solver totals including cache reads.